

Task 12: Log Monitoring and Analysis

CHAPTER 1: Introduction to Log Monitoring

Log monitoring is the process of collecting, reviewing, and analyzing system-generated logs to understand system behavior, detect security incidents, and troubleshoot issues. Logs act as a digital record of activities performed by users, applications, and system services. In cybersecurity, log monitoring is an essential defensive mechanism used to identify unauthorized access, misconfigurations, and suspicious activities.

Effective log monitoring provides visibility into system operations and enables early detection of potential threats, helping organizations respond to incidents in a timely manner.

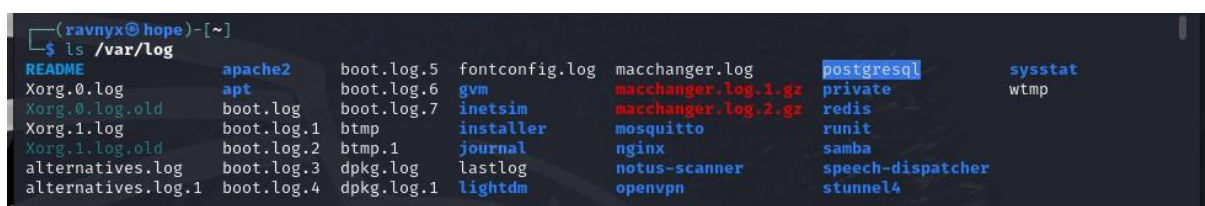
CHAPTER 2: Types of Logs in Linux Systems

Linux systems generate various types of logs, each serving a specific purpose. Understanding these log types is essential for effective monitoring and analysis. Common categories include:

- **Authentication Logs:** Record login attempts, sudo usage, and authorization events.
- **System Logs:** Capture system startup, shutdown, and service-related activities.
- **Application Logs:** Store logs generated by applications such as web servers and databases.
- **Kernel Logs:** Contain messages related to hardware and kernel-level operations.

Each log type provides different insights into system behavior and security events.

Figure 2.1: Listing of System Log Directories in Linux



```
(ravnyx@hope)~  
$ ls /var/log  
README          apache2          boot.log.5      fontconfig.log  macchanger.log  postgresql      sysstat  
Xorg.0.log       apt              boot.log.6      gvm             macchanger.log.1.gz private         wtmp  
Xorg.0.log.old   boot.log         boot.log.7      inetsim         macchanger.log.2.gz redis  
Xorg.1.log       boot.log.1      bttmp           installer        mosquito        runit  
Xorg.1.log.old   boot.log.2      bttmp.1         journal         nginx           samba  
alternatives.log boot.log.3      dpkg.log        lastlog         notus-scanner   speech-dispatcher  
alternatives.log.1 boot.log.4      dpkg.log.1      lightdm         openvpn         stunnel4
```

(This screenshot displays various system, authentication, and application log directories available in Kali Linux for monitoring and analysis.)

CHAPTER 3: Authentication Log Analysis

Authentication logs are critical for understanding user access behavior and privilege usage. These logs record successful and failed login attempts, sudo command executions, and session activity. By analyzing authentication logs, security analysts can detect unauthorized access attempts and misuse of elevated privileges.

In this task, authentication-related activities were analyzed using system logging utilities to observe login sessions, SSH activity, and sudo operations.

Figure 3.1: SSH Authentication Events Captured Using systemd Journal

```
(ravnyx@hope)-[~]
$ sudo journalctl -u ssh
Jan 29 13:13:53 hope systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Jan 29 13:13:53 hope sshd[5864]: Server listening on 0.0.0.0 port 22.
Jan 29 13:13:53 hope sshd[5864]: Server listening on :: port 22.
Jan 29 13:13:53 hope systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Jan 29 13:16:41 hope sshd-session[7234]: Connection closed by 127.0.0.1 port 57112
Jan 29 13:21:34 hope sshd-session[9640]: Connection closed by 127.0.0.1 port 39866
Jan 29 13:21:52 hope sshd-session[9788]: error: Protocol major versions differ: 2 vs. 1
Jan 29 13:21:52 hope sshd-session[9788]: banner exchange: Connection from 127.0.0.1 port 37670: could not read prot>
Jan 29 13:21:52 hope sshd-session[9789]: Unable to negotiate with 127.0.0.1 port 37672: no matching host key type f>
Jan 29 13:21:52 hope sshd-session[9793]: Connection closed by 127.0.0.1 port 37686 [preauth]
Jan 29 13:21:52 hope sshd-session[9795]: Unable to negotiate with 127.0.0.1 port 37696: no matching host key type f>
Jan 29 13:21:52 hope sshd-session[9797]: Unable to negotiate with 127.0.0.1 port 37702: no matching host key type f>
Jan 29 13:21:52 hope sshd-session[9799]: Connection closed by 127.0.0.1 port 37710 [preauth]
Jan 29 15:35:42 hope systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server ...
Jan 29 15:35:42 hope sshd[5864]: Received signal 15; terminating.
Jan 29 15:35:42 hope systemd[1]: ssh.service: Deactivated successfully.
Jan 29 15:35:42 hope systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
```

(SSH-related authentication events including service start, stop, and connection attempts were analyzed using journalctl.)

CHAPTER 4: Identification of Failed Login Attempts

Failed login attempts are important indicators of potential security threats such as brute-force attacks or unauthorized access attempts. These failures may occur due to incorrect credentials, unknown users, or service misconfigurations.

Figure 4.1: Detection of Failed Authentication Attempts in System Logs

```

(ravnix@hope)-[~]
$ sudo journalctl | grep -i "auth"
Apr 28 13:30:54 hope systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RPCSEC_GSS was skipped because of an unmet condition check (ConditionPathExists=/etc/krb5.keytab).
Apr 28 13:30:56 hope systemd[1]: Starting polkit.service - Authorization Manager...
Apr 28 13:30:57 hope systemd[1]: Started polkit.service - Authorization Manager.
Apr 28 13:31:18 hope polkitd[556]: Registered Authentication Agent for unix-session:2 (system bus name :1.44 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8)
Apr 28 14:05:12 hope systemd[1]: Stopping polkit.service - Authorization Manager...
Apr 28 14:05:12 hope systemd[1]: Stopped polkit.service - Authorization Manager.
Apr 28 14:05:12 hope systemd[1]: Starting polkit.service - Authorization Manager...
Apr 28 14:05:13 hope systemd[1]: Started polkit.service - Authorization Manager.
Apr 28 14:05:13 hope polkitd[63486]: Registered Authentication Agent for unix-session:2 (system bus name :1.44 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8)
Jun 01 13:27:46 hope systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RPCSEC_GSS was skipped because of an unmet condition check (ConditionPathExists=/etc/krb5.keytab).
Jun 01 13:27:47 hope systemd[1]: Starting polkit.service - Authorization Manager...
Jun 01 13:27:48 hope systemd[1]: Started polkit.service - Authorization Manager.
Jun 01 13:28:14 hope lightdm[901]: pam_unix(lightdm:auth): check pass; user unknown
Jun 01 13:28:14 hope lightdm[901]: pam_unix(lightdm:auth): authentication failure; logname= uid=0 euid=0 tty=:0 ruse r= rhost=
Jun 01 13:29:29 hope lightdm[903]: pam_unix(lightdm:auth): check pass; user unknown
Jun 01 13:29:29 hope lightdm[903]: pam_unix(lightdm:auth): authentication failure; logname= uid=0 euid=0 tty=:0 ruse r= rhost=
Jun 01 13:29:52 hope polkitd[564]: Registered Authentication Agent for unix-session:2 (system bus name :1.47 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8)
Jun 01 15:28:15 hope polkitd[564]: Unregistered Authentication Agent for unix-session:2 (system bus name :1.47, object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8) (disconnected from bus)
Jun 01 15:28:15 hope systemd[1]: Stopping polkit.service - Authorization Manager...
Jun 01 15:28:15 hope systemd[1]: Stopped polkit.service - Authorization Manager.
Jan 16 11:42:05 hope systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RPCSEC_GSS was skipped because of an unmet condition check (ConditionPathExists=/etc/krb5.keytab).
Jan 16 11:42:06 hope systemd[1]: Starting polkit.service - Authorization Manager...
Jan 16 11:42:07 hope systemd[1]: Started polkit.service - Authorization Manager.
Jan 16 11:42:31 hope polkitd[538]: Registered Authentication Agent for unix-session:2 (system bus name :1.47 [/usr/libexec/polkit-mate-authentication-agent-1], object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8)
Jan 16 18:48:35 hope polkitd[538]: Unregistered Authentication Agent for unix-session:2 (system bus name :1.47, object path /org/mate/PolicyKit1/AuthenticationAgent, locale C.UTF-8) (disconnected from bus)
Jan 16 18:48:35 hope systemd[1]: Stopping polkit.service - Authorization Manager...
Jan 16 18:48:35 hope systemd[1]: Stopped polkit.service - Authorization Manager.
Jan 19 13:17:49 hope systemd[1]: auth-rpcgss-module.service - Kernel Module supporting RPCSEC_GSS was skipped because of an unmet condition check (ConditionPathExists=/etc/krb5.keytab).

```

(Failed login attempts and authentication errors were identified through system log analysis, indicating unsuccessful access attempts.)

By monitoring failed authentication events in logs, administrators can quickly identify suspicious behavior and take appropriate security measures. The analysis demonstrated how failed login attempts can be detected through log review.

CHAPTER 5: Anomaly Detection in Logs

Anomaly detection involves identifying activities that deviate from normal system behavior. Examples include repeated authentication failures, frequent sudo usage, unusual service restarts, or unexpected protocol errors.

Figure 5.1: Anomalous Authentication and Privilege Escalation Activity


```

Jan 16 11:43:31 hope sudo[2062]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=0)
Jan 16 11:43:58 hope sudo[2062]: pam_unix(sudo:session): session closed for user root
Jan 16 18:48:35 hope sudo[1477]: pam_unix(sudo:session): session closed for user root
Jan 19 13:19:33 hope sudo[1632]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/su
Jan 19 13:19:33 hope sudo[1632]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Jan 19 13:58:09 hope sudo[1632]: pam_unix(sudo:session): session closed for user root
Jan 19 16:19:51 hope sudo[4468]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/su
Jan 19 16:19:51 hope sudo[4468]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Jan 19 16:20:27 hope sudo[4885]: pam_unix(sudo:auth): conversation failed
Jan 19 16:20:27 hope sudo[4885]: pam_unix(sudo:auth): auth could not identify password for [ravnyx]
Jan 19 16:39:55 hope sudo[4468]: pam_unix(sudo:session): session closed for user root
Jan 20 11:11:20 hope sudo[6956]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/gunzip /usr/
share/wordlists/rockyou.txt.gz
Jan 20 11:11:20 hope sudo[6956]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Jan 20 11:11:22 hope sudo[6956]: pam_unix(sudo:session): session closed for user root
Jan 26 13:42:57 hope sudo[25300]:    ravnyx : TTY=pts/1 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/systemctl s
tart apache2
Jan 26 13:42:57 hope sudo[25300]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Jan 26 13:42:57 hope sudo[25300]: pam_unix(sudo:session): session closed for user root
Jan 26 13:43:27 hope sudo[25626]:    ravnyx : TTY=pts/1 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/systemctl s
tart mysql

```

(Repeated sudo sessions and authentication events observed within a short time frame indicate anomalous behavior patterns.)

Anomalies are not always malicious, but they can indicate security misconfigurations or early signs of an attack. Log analysis helps in recognizing such irregular patterns and supports proactive monitoring.

CHAPTER 6: Event Correlation

Event correlation is the process of linking related log events to understand the full context of system activity. By correlating authentication attempts with service start and stop events, analysts can reconstruct event timelines and identify relationships between different system actions.

This approach helps in distinguishing isolated events from coordinated or suspicious activity.

Figure 6.1: Correlation of SSH Service Lifecycle and Authentication Events

```

(ravnyx@hope)-[~]
$ sudo journalctl -n 20
Feb 03 15:20:05 hope sudo[2117]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/cat /var/lo
Feb 03 15:20:05 hope sudo[2117]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:20:05 hope sudo[2117]: pam_unix(sudo:session): session closed for user root
Feb 03 15:20:27 hope sudo[2367]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/less /var/l
Feb 03 15:20:27 hope sudo[2367]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:20:27 hope sudo[2367]: pam_unix(sudo:session): session closed for user root
Feb 03 15:21:41 hope sudo[2950]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 15:21:41 hope sudo[2950]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:23:28 hope sudo[2950]: pam_unix(sudo:session): session closed for user root
Feb 03 15:23:48 hope sudo[3960]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 15:23:48 hope sudo[3960]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:23:48 hope sudo[3960]: pam_unix(sudo:session): session closed for user root
Feb 03 15:24:06 hope sudo[4124]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 15:24:06 hope sudo[4124]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:24:06 hope sudo[4124]: pam_unix(sudo:session): session closed for user root
Feb 03 15:24:31 hope sudo[4345]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 15:24:31 hope sudo[4345]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)
Feb 03 15:24:31 hope sudo[4345]: pam_unix(sudo:session): session closed for user root
Feb 03 15:24:45 hope sudo[4481]:    ravnyx : TTY=pts/0 ; PWD=/home/ravnyx ; USER=root ; COMMAND=/usr/bin/journalctl
Feb 03 15:24:45 hope sudo[4481]: pam_unix(sudo:session): session opened for user root(uid=0) by ravnyx(uid=1000)

```

(SSH service lifecycle events were correlated with authentication attempts using timestamps to understand event relationships.)

CHAPTER 7: SIEM (Security Information and Event Management) Basics

SIEM systems are designed to collect logs from multiple sources, normalize data, correlate events, and generate alerts for suspicious activities. SIEM tools enhance visibility by centralizing log analysis and enabling real-time monitoring.

Common SIEM Tools and Their Role

Security Information and Event Management (SIEM) tools are widely used in organizations to centralize log collection, correlate events, and generate security alerts. Some commonly used SIEM tools include:

- **Splunk** – Used for log aggregation, search, visualization, and alerting
- **IBM QRadar** – Focuses on real-time threat detection and correlation
- **Elastic Stack (ELK)** – Used for log collection, indexing, and visualization
- **ArcSight** – Enterprise SIEM for large-scale security monitoring

These tools automate log analysis by correlating events from multiple sources, reducing manual effort and improving incident detection. In this task, SIEM concepts were studied theoretically to understand how automated monitoring systems enhance security operations.

Although no SIEM tool was configured in this task, basic SIEM concepts were studied theoretically to understand centralized log analysis and alerting mechanisms used in enterprise environments.

CHAPTER 8: Alerting and Incident Awareness

Alerts notify administrators when predefined conditions are met, such as multiple failed login attempts or repeated privilege escalation events. Effective alerting reduces response time and helps prevent potential security incidents from escalating.

Sample Security Alert Conditions

- Alert when **more than 5 failed login attempts** occur within a short time window
- Alert on **repeated sudo privilege escalation attempts**
- Alert when **SSH service restarts unexpectedly**
- Alert on **authentication attempts from unknown users**

Understanding alert conditions is essential for designing proactive security monitoring strategies.

CHAPTER 9: Documentation of Findings

Documenting log analysis findings ensures transparency and provides a reference for future investigations. Proper documentation includes observed events, detected anomalies, correlated activities, and security implications. (Findings from log analysis were documented based on observed authentication events, anomalies, and correlated system activity.)

This step is essential for incident reporting, audits, and compliance requirements.

CHAPTER 10: Security Importance of Log Monitoring

Log monitoring plays a critical role in maintaining the security and stability of computer systems. Logs provide detailed records of user activities, authentication attempts, system events, and service operations. Continuous log monitoring enables early detection of unauthorized access, privilege misuse, and suspicious behavior.

Logs also serve as forensic evidence during incident response and security investigations. They help trace the sequence of events leading to an incident and assess the scope of potential security breaches.

CHAPTER 11: Limitations of Manual Log Analysis

While manual log analysis is effective for learning and small-scale environments, it has limitations in real-world systems. Large infrastructures generate massive volumes of logs, making manual review time-consuming and prone to human error.

To address these challenges, organizations rely on automated monitoring tools and SIEM solutions that provide real-time analysis, correlation, and alerting capabilities.